Reproduction Plan for Figueras–Andrade–França–Gu: 5D Gregory–Laflamme Black String Dynamics

Purpose

The immediate goal is not yet to compute gravitational radiation. The first goal is to reproduce the diagnostic backbone of the 5D black-string evolution in arXiv:2210.13501. This document is written as a working plan for a first-principles reproduction, starting from simulation setup, apparent-horizon extraction, and post-processing.

The six target outputs are:

1. Apparent-horizon area versus time.
2. Minimum string radius versus time.
3. Logarithmic pinch-off scaling.
4. Fourier coefficients of the horizon radius on string segments.
5. Generation formation times and geometric properties.
6. Convergence of all of the above.

The practical philosophy is: first reproduce the cleanest case, `L/r0 = 10`, then repeat for `L/r0 = 8, 12, 16`, and only then begin changing physics or adding radiation extraction.

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0. Shared Baseline Setup

0.1 Physical system

Work in 4+1-dimensional vacuum GR with one compact string direction `z ~ z + L`. The target initial geometry is a uniform 5D black string with Schwarzschild mass parameter `r0`, plus a small GL-triggering perturbation localized near the horizon.

Use units

```
G = c = 1,
r0 = 1 for production reproduction runs.
```

The primary reproduction cases are:

```
L/r0 = 8, 10, 12, 16.
```

The most important benchmark is `L/r0 = 10`, because that case is the closest direct comparison with the classic Lehner–Pretorius result and with the paper being reproduced.

0.2 Evolution formulation and gauge

Use the same numerical choices as closely as possible before introducing improvements:

```
Formulation: 4+1 CCZ4.
Symmetry reduction: SO(3) modified cartoon, reducing the effective problem to
2+1.
Coordinates: Cartesian/radial cartoon coordinate x plus compact coordinate z.
Spatial derivatives: 6th-order finite differences.
Time integration: RK4.
Dissipation: 6th-order Kreiss–Oliger for direct reproduction of the paper.
Gauge: 1+log lapse and integrated Gamma-driver shift.
```

Note on dissipation: for a faithful reproduction, use the paper's 6th-order Kreiss–Oliger choice first. Do not change to 8th-order KO in the baseline reproduction run, even if one later wants to test higher-order dissipation as a numerical-systematics check. The paper explicitly chose 6th-order KO because RK4 and AMR already prevent the full 6th-order spatial discretization from being realized globally.

Gauge and damping parameters to reproduce first:

```
kappa1 = 0.37, with kappa1 -> kappa1/alpha,
kappa2 = -0.8,
c_alpha = 1.3,
c_beta = 0.6.
```

Initial lapse and shift:

```
alpha(t=0) = 1,
beta^i(t=0) = 0.
```

0.3 Initial data

Start from the 5D black string in Gullstrand–Painlevé coordinates, then perturb the conformal factor. The perturbation should be localized near the horizon and should not change the ADM mass asymptotically.

Use the conformal-factor perturbation

```
chi = 1 + epsilon * sin(2 pi n z / L) * exp[-(x/r0 - r0/x)^2].
```

Default reproduction values:

```
epsilon = 0.01,
n = 1,
r0 = 1.
```

Use a turduckening-style regularization of the initial singularity:

```
if x < eps_cut, evaluate initial-data fields at x = eps_cut,
eps_cut = 0.1 r0.
```

Important implementation point: the perturbation intentionally introduces small constraint violations. The first validation step is to verify that CCZ4 damping suppresses these violations on a timescale much shorter than the GL growth timescale.

0.4 Domain, boundary conditions, AMR

Use

```
x_outer = 256 r0,
z periodic with period L.
```

The outer boundary in the `x` direction can use Sommerfeld or periodic boundary conditions, provided it is not in causal contact with the black string during the relevant evolution.

Baseline grid:

```
coarsest spacing h = 0.25 r0,
13 refinement levels total,
2:1 refinement ratio.
```

The important reproduction detail is the AMR strategy. Use rectangular refinement boxes rather than fully dynamic gradient-based tagging. The goal is not just efficient resolution, but preservation of the physical `Z2` symmetry about the center of the relevant string segment. Dynamic refinement based only on `|partial chi|` can introduce asymmetric refinement boundaries, which can seed unphysical late-time effects.

Operational rule:

```
Require at least 40 grid points across the thinnest apparent-horizon radius.
Aim for 60+ points when feasible.
```

```
When this fails, add a refinement level.
Once a rectangular refinement level is added, avoid frequent regridding.
```

0.5 Horizon representation

Represent the apparent horizon at each output time as a graph

```
x = h(t, z),    z in [0, L),
```

with the `S^2` directions reconstructed by SO(3) symmetry.

For each horizon time slice, save the following data:

```
t,
z_j,
h(t,z_j),
metric components on the horizon: gamma_xx, gamma_xz, gamma_zz, gamma_ww,
areal radius R(t,z_j),
proper-length integrand along z,
optional: Kretschmann scalar on the horizon,
AMR level at each horizon point,
constraint norms near the horizon.
```

The areal radius of the horizon two-sphere is

```
R(t,z) = sqrt(gamma_ww(t, h(t,z), z)) * h(t,z),
```

where `w` denotes one of the Cartesian cartoon directions along the `S^2`.

The proper-length element along the horizon curve is

```
dl_z = sqrt(gamma_zz + 2 gamma_xz h_z + gamma_xx h_z^2) dz.
```

The apparent-horizon area is then

```
A(t) = 4 pi integral_0^L R(t,z)^2 dl_z.
```

The unperturbed uniform black-string area is

```
A0 = 4 pi r0^2 L.
```

All area plots should use `A/A0`; all time axes should use `t/r0`; all radii should use `R/r0`.

0.6 Data products

Create a consistent run directory structure:

```
runs/
  L10_med/
    checkpoints/
    horizon_raw/
    horizon_processed/
    diagnostics/
    plots/
    logs/
  L10_low/
  L10_high/
  L8_med/
  L12_med/
  L16_med/
```

Minimum output cadence:

```
Early gauge settling, 0 <= t/r0 <= 20: moderate cadence.
Linear GL growth: moderate cadence.
Nonlinear first generation: high cadence.
Late cascade: very high cadence, because event times and R_min become sensitive.
```

A useful target is to save horizon diagnostics at least every

```
Delta t/r0 ~ 0.1
```

near the nonlinear cascade, and more frequently if storage allows.

---

1. Apparent-Horizon Area Versus Time

1.1 Target

Reproduce the plot of normalized apparent-horizon area

```
A(t)/A0 versus t/r0
```

for `L/r0 = 8, 10, 12, 16`, with special focus on `L/r0 = 10`.

The qualitative benchmark is:

- `A/A0` starts near 1.
- There is a short early gauge-adjustment transient.
- The area then grows during nonlinear GL evolution.
- For `L=10`, the final simulated value should be close to `A/A0 ~ 1.37` before the run terminates.
- Larger `L` cases should reach larger final normalized areas.
- The curve shape should match the paper's Figure 2 at the level of timing and overall plateau values, modulo small differences caused by gauge and AMR details.

1.2 Required inputs

For every horizon output time:

```
h(t,z),
h_z(t,z),
gamma_xx, gamma_xz, gamma_zz, gamma_ww on the horizon,
L, r0.
```

1.3 Algorithm

For each time slice:

1. Interpolate metric components from the numerical grid to the apparent-horizon surface `x = h(t,z)`.
2. Compute the areal radius:

```
R(t,z) = sqrt(gamma_ww) * h(t,z).
```

1. Compute `h_z` using either:

```
spectral/periodic differentiation in z,
```

or a high-order finite-difference derivative consistent with the horizon grid.

1. Compute the proper-length element:

```
dl_z = sqrt(gamma_zz + 2 gamma_xz h_z + gamma_xx h_z^2) dz.
```

1. Compute the area:

```
A(t) = 4 pi * integral R(t,z)^2 dl_z.
```

1. Normalize:

```
A_norm(t) = A(t) / (4 pi r0^2 L).
```

1. Store:

```
t/r0, A/A0, estimated integration error, horizon finder residual.
```

1.4 Numerical details

Use a periodic integration rule in `z`, such as the trapezoidal rule on a uniform periodic grid. For smooth periodic data this is highly accurate. If the horizon points are not uniform in `z`, resample to a uniform periodic grid before differentiation and integration.

Implement two independent area estimators:

```
Estimator A: direct quadrature over raw horizon points.
Estimator B: resampled periodic spline/Fourier interpolation, then quadrature.
```

The two estimates should agree well before trusting late-time area values.

1.5 Plots

Produce the following:

1. Single-case plot:

```
A/A0 versus t/r0 for L=10.
```

1. Multi-case plot:

```
A/A0 versus t/r0 for L=8,10,12,16.
```

1. Diagnostic plot:

```
absolute difference between area estimators A and B versus t/r0.
```

1. Optional reference overlays:

```
asymptotically flat 5D Schwarzschild area with same total mass,
approximate KK black-hole area with same mass,
approximate two-KK-black-hole area using first/second generation radii.
```

The optional overlays are not necessary for first reproduction, but they are useful for checking the thermodynamic interpretation.

1.6 Validation criteria

For `L=10`, require:

```
final A/A0 near 1.37,
area growth onset around the same time as the paper,
curve shape agrees visually with Figure 2,
area remains monotonic up to small numerical noise.
```

For all `L`, require:

```
A/A0 starts at 1 within perturbative error,
no unphysical area drops larger than numerical error,
qualitative ordering of final area values agrees with the paper.
```

1.7 Failure modes

Likely problems and fixes:

- **Area decreases strongly:** horizon finder is failing, interpolation is noisy, or the time cadence is too sparse.
- **Area grows during the gauge-adjustment stage too much:** initial gauge dynamics or constraint violations are not under control.
- **Late-time oscillatory area:** AMR boundaries or horizon reconstruction are introducing noise.
- **Different final plateau:** check normalization by `A0 = 4 pi r0^2 L`, units of `r0`, and whether the simulation has actually reached comparable late time.

---

2. Minimum String Radius Versus Time

2.1 Target

Reproduce the time evolution of the thinnest apparent-horizon radius,

```
R_min(t)/r0 versus t/r0,
```

where

```
R_min(t) = min_z R(t,z).
```

This is the raw diagnostic underlying the pinch-off scaling analysis.

2.2 Required inputs

For every apparent-horizon output time:

```
t,
z_j,
R(t,z_j),
horizon finder residual,
AMR level at horizon points.
```

2.3 Algorithm

For each time:

1. Compute the areal radius `R(t,z)` as in the area diagnostic.
2. Smooth only if necessary, using a conservative periodic filter. Keep both raw and smoothed versions.
3. Find all local minima and maxima of `R(t,z)` on the circle.
4. Define:

```
R_min_raw(t) = minimum over raw R(t,z),
z_min_raw(t) = location of that minimum,
R_min_smooth(t) = minimum over smoothed R(t,z),
z_min_smooth(t) = location of that minimum.
```

1. Use the raw value for final reported data if it is stable. Use smoothed values only as a diagnostic or if raw pointwise noise is clearly contaminating the minimum.

2. Track the location `z_min(t)` continuously. Sudden jumps can be physical if the thinnest segment changes, but they can also reveal insufficient resolution.

2.4 Distinguish physical minima from numerical minima

At late times, the minimum is often near a bulge/string junction or inside a small string segment. Therefore:

- Do not assume the minimum lies at a fixed symmetry point.
- Track every local minimum, not just the global minimum.
- Store the full ordered list:

```
{z_min,k(t), R_min,k(t)}.
```

- Identify which local minimum becomes the global minimum as the cascade progresses.

2.5 Plots

Produce:

1. Linear plot:

```
R_min/r0 versus t/r0.
```

1. Semilog plot:

```
log(R_min/r0) versus t/r0.
```

1. Location plot:

```
z_min/r0 versus t/r0.
```

1. Multi-minima plot:

```
first few local minima R_min,k/r0 versus t/r0.
```

The location plot is important because a clean pinch-off interpretation requires knowing whether the same physical segment is being followed.

2.6 Validation criteria

For a successful reproduction:

```
R_min decreases rapidly after nonlinear GL growth begins.
R_min approaches zero approximately linearly in tc - t during late evolution.
The late-time minimum remains well resolved by the finest AMR level.
The location of R_min is consistent with the horizon embedding diagrams.
```

For `L=10`, the late-time behavior should be compatible with a pinch-off time near

```
tc/r0 ~ 128.
```

Do not use this number as an input to the simulation; use it only as a posterior validation.

2.7 Failure modes

- **Minimum is a single-grid-point spike:** insufficient horizon resolution or noisy interpolation.
- **Minimum jumps randomly between unrelated locations:** noisy extrema finder; increase smoothing or output cadence.
- **Minimum stalls at finite value:** not enough refinement levels, time step too large, or simulation terminates before entering asymptotic regime.
- **Minimum becomes negative or undefined:** areal-radius reconstruction is wrong or the horizon finder failed.

---

3. Logarithmic Pinch-Off Scaling

3.1 Target

Reproduce the approximate linear pinch-off scaling

```
R_min(t) = a (tc - t)
```

and the log-log plot

```
log(R_min/r0) versus log((tc - t)/r0).
```

The expected slope in the log-log plot is approximately 1. The coefficient `a` is small, around a few times `10^{-3}` in the paper's normalization.

3.2 Inputs

Use the time series from Item 2:

```
t_i,
R_min(t_i),
error estimate for R_min(t_i),
z_min(t_i),
resolution metadata.
```

3.3 Fitting strategy

Use several fitting strategies, not just one, because `tc` and `a` are correlated.

Fit A: direct two-parameter late-time fit

Fit

```
R_min(t) = a (tc - t)
```

using nonlinear least squares over a chosen late-time window.

Parameters:

```
a,
tc.
```

Constraints:

```
a > 0,
tc > max(t_i).
```

Fit B: three-parameter power-law fit

Fit

```
R_min(t) = b (tc - t)^p.
```

Parameters:

```
b,
tc,
p.
```

This checks whether the exponent is actually consistent with `p = 1`.

Fit C: fixed-exponent scan

For a grid of candidate `tc`, define

```
y_i = R_min(t_i),
x_i(tc) = tc - t_i.
```

For each `tc`, fit a straight line through the origin:

```
y_i = a x_i.
```

Pick the `tc` minimizing the residual. This is robust and transparent.

3.4 Fitting windows

The dominant systematic error is the choice of late-time window. Use multiple windows:

```
Window 1: after first-generation formation.
Window 2: after second-generation formation.
Window 3: only latest 30% of available R_min data.
Window 4: only points satisfying R_min/r0 < threshold.
```

Example thresholds:

```
R_min/r0 < 0.2,
R_min/r0 < 0.1,
R_min/r0 < 0.05.
```

Report the spread over windows as a systematic error.

3.5 Log-log plot construction

After choosing a best-fit `tc`, plot:

```
x-axis: -log((tc - t)/r0) or log((tc - t)/r0), matching the paper’s convention.
y-axis: log(R_min/r0).
```

Also plot the equatorial radii of several generations on the same axes:

```
R_1(t), R_2(t), R_3(t), ...
```

This reproduces the structure of the paper's Figure 10.

3.6 Validation criteria

For `L=10`, expect approximately:

```
tc/r0 ~ 128,
a ~ 0.005.
```

For the other `L` values, expect the coefficient `a` to remain of the same order and fairly insensitive to `L`, while `tc` changes strongly with `L` because the nonlinear onset time changes.

A successful result should show:

```
log-log slope close to 1,
stable tc under reasonable late-time fitting-window changes,
R_min controlled by a physically tracked local minimum rather than numerical
noise.
```

3.7 Failure modes

- **Exponent not near 1:** fit window is too early, or `tc` estimate is unstable.
- **Large changes in `tc` under small window changes:** simulation has not reached the scaling regime.
- **Late-time points bend away from the line:** horizon resolution is failing, or AMR/time step is inadequate near pinch-off.
- **Different `a` by factors of several:** check normalization of `R`, `t`, and `r0`; also check whether you used `R_min = min R` rather than coordinate radius `h`.

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4. Fourier Coefficients of the Horizon Radius on String Segments

4.1 Target

Reproduce the Fourier analysis of the apparent-horizon radius on an unstable string segment. The core diagnostic is the relative strength of GL harmonics:

```
c_k(t) / c_0(t).
```

For the first-generation string segment in the `L=10` case, the expected result is:

```
c_1 dominates,
even modes are negligible,
c_3 and c_5 are much smaller than c_1 during the relevant interval.
```

For larger `L`, higher odd modes should become more important relative to the fundamental mode.

4.2 Inputs

For each time:

```
R(t,z),
local extrema of R(t,z),
identified bulge centers,
identified string-segment endpoints,
proper or coordinate segment length,
choice of segment coordinate.
```

4.3 Segment definition

There are two different goals that should not be mixed:

```
Goal A: reproduce the paper's Figure 6 exactly.
Goal B: construct a more invariant and better-conditioned harmonic diagnostic.
```

For **Goal A**, start with the paper's convention:

```
L=10,
first-generation string segment,
coordinate region z in [0, 5 r0], or the translated/periodic equivalent that
covers the same first-generation segment.
```

This fixed coordinate interval is not an arbitrary choice; it is the convention used in the paper for the L=10 Fourier plot. Therefore it should remain the primary reproduction diagnostic.

For **Goal B**, define the segment dynamically:

```
Identify neighboring extrema of R(t,z) that bound the parent string segment.
Map the selected segment to a local coordinate s in [0, L_seg].
Optionally replace coordinate length by proper horizon length.
```

This dynamic segment definition is useful as a cross-check and will likely be preferable for later radiation-source diagnostics, but it should not replace the paper convention when reproducing Figure 6.

4.4 Basis and coefficient extraction

For **direct reproduction of the paper**, use the sine decomposition used in their Eq. (11):

```
R(t,z) = c0(t) + sum_{k=1}^{Kmax} c_k(t) sin(pi k z / L_s,1).
```

This is the convention needed to reproduce their plotted ratios `c1/c0`, `c3/c0`, and `c5/c0`.

For a **basis-systematics cross-check**, also compute a cosine expansion on a dynamically identified segment:

```
R(t,s) = a0(t)/2 + sum_{k=1}^{Kmax} a_k(t) cos(pi k s / L_seg),
```

where `s in [0, L_seg]` is measured from one dynamically identified endpoint of the string segment. The cosine basis is more natural if the chosen endpoints are extrema of `R`, since then `partial_s R = 0` at the endpoints. This should be treated as a robustness/invariant-diagnostic check, not as the primary reproduction convention.

Practical implementation:

1. For the paper diagnostic, use the fixed coordinate segment and sine basis.
2. For the robustness diagnostic, map the dynamically identified segment to `s in [0, L_seg]` and use the cosine basis.
3. Interpolate `R(t,s)` or `R(t,z)` to a uniform grid on the chosen interval.
4. Choose `Kmax`, for example:

```
Kmax = 10 for first pass,
Kmax = 20 for robustness checks.
```

1. Solve for coefficients by discrete least squares rather than relying on analytic orthogonality formulas, because the numerical segment conventions and endpoint values can differ slightly.
2. Store:

```
sine-basis c0(t), c1(t), ..., cKmax(t),
cosine-basis a0(t), a1(t), ..., aKmax(t),
```

```
c_k/c0 and a_k/a0,
least-squares residuals,
segment endpoints,
coordinate and proper segment lengths.
```

4.5 Time interval

For `L=10`, focus on the interval around the formation of the second generation, roughly:

```
t/r0 ~ 100 to 117
```

The exact interval should be determined from the horizon data. The lower bound should be after the first-generation segment is identifiable; the upper bound should be before the second generation has grown so much that the original single-segment decomposition becomes ambiguous.

4.6 Plots

Produce:

1. Primary reproduction plot:

```
c1/c0, c3/c0, c5/c0 versus t/r0
```

for the first-generation segment in the `L=10` run.

1. Even-mode diagnostic:

```
c2/c0, c4/c0, c6/c0 versus t/r0.
```

1. Residual diagnostic:

```
||R - R_fit|| / ||R|| versus t/r0.
```

1. Multi-`L` comparison:

```
c1/c0, c3/c0, c5/c0 for L=8,10,12,16.
```

4.7 Validation criteria

For `L=10`:

```
c1/c0 dominates the first-generation segment dynamics,
c3/c0 remains much smaller until late in the interval,
c5/c0 is smaller still,
even coefficients are close to zero compared with odd coefficients.
```

For larger `L`:

```
higher odd modes become visibly stronger,
consistent with longer string segments supporting more unstable harmonics.
```

4.8 Failure modes

- **Even modes large:** segment not centered symmetrically, AMR broke reflection symmetry, or endpoint convention is wrong.
- **c1 does not dominate for L=10:** wrong segment selected, wrong time interval, or wrong basis length.
- **Large residuals:** the segment is too nonlinear for the chosen Fourier expansion, or the endpoints include bulges rather than the string segment.
- **Coefficients jump discontinuously:** segment tracking is failing; use extrema tracking to define continuous endpoints.

---

5. Generation Formation Times and Geometric Properties

5.1 Target

Reproduce tables of generation properties, especially for `L=10`, and then for `L=8,12,16`.

The quantities to measure are:

```
generation index i,
formation time,
number of satellite bulges per string segment n_s,
string-segment radius R_s,i,
bulge/equatorial radius R_h,i or R_h,f,
string-segment length L_s,i,
ratios such as L_s,i/R_s,i or R_s,i/L_s,i.
```

There are two relevant formation-time definitions:

1. Lehner–Pretorius-style threshold definition.
2. The derivative-based definition used for the paper's Table II.

Implement both.

5.2 Horizon feature extraction

At each time, start from `R(t,z)`.

1. Smooth gently in `z` if necessary.
2. Find all local maxima and minima.
3. Classify:

```
local maxima -> candidate bulges,
local minima or plateaus -> candidate string segments/necks.
```

1. Track extrema across time using nearest-neighbor or Hungarian matching in periodic `z`:

```
match extrema at t_i to extrema at t_{i+1}
using distance in z and radius continuity.
```

1. Create tracks:

`text`
`   bulge_track_id,`
`z_bulge(t),`
`R_bulge(t),`
`neighboring minima,`
`parent segment.`

1. Store events when new local maxima appear and persist longer than a minimum duration.

5.3 Definition 1: threshold formation time

This definition should be used only to reproduce the Lehner–Pretorius-style comparison table, not as the main quantitative definition of formation time.

For each new bulge candidate, define its surrounding string radius `R_surr(t)`. Options:

```
R_surr = average of the two neighboring local minima,
R_surr = minimum of the two neighboring minima,
R_surr = local plateau fit excluding the bulge.
```

Use one convention as primary and the others as error estimates.

The threshold formation time is the first time satisfying

```
R_bulge(t) >= 1.5 R_surr(t).
```

At that time, measure:

```
t_i,
n_s,
R_s,i,
R_h,i,
L_s,i.
```

This threshold is gauge/slicing/convention dependent and should be described as a heuristic event marker. It is retained because it allows direct comparison with the older Lehner–Pretorius-style table.

5.4 Definition 2: derivative-based formation time

This is the primary quantitative definition for the reproduction, because it is the one used for the paper's multi-L generation table.

For each bulge track, compute the logarithmic growth rate:

```
d ln R_h,i / dt.
```

Use a smoothing differentiator such as:

```
Savitzky–Golay filter,
local polynomial regression,
or spline derivative with cross-validation.
```

Then define:

```
t_p,i = time of the absolute maximum of d ln R_h,i/dt,
t_n,i = time of the first subsequent local minimum of d ln R_h,i/dt.
```

The paper uses `t_n,i` as the formation time for measuring geometric properties.

At `t_n,i`, measure:

```
R_s,i,
R_h,f,
L_s,i,
n_s.
```

5.5 Measuring geometric properties

Bulge radius

For a bulge maximum at `z_b`, define

```
R_h,i = R(t, z_b).
```

If the maximum is broad or noisy, fit a local quadratic or quartic polynomial near the maximum and use the fitted peak.

String radius

For the string segment associated with generation `i`, define `R_s,i` as a representative local minimum/neck radius.

Because the string becomes nonuniform at late generations, report several choices:

```
R_s,min = minimum neck radius in the segment,
R_s,avg = average over the flattest central portion of the segment,
R_s,parent = parent segment radius just before bulge formation.
```

Use one as the primary convention, but store all three.

Segment length

Measure both:

```
coordinate length in z,
proper length along the horizon curve.
```

For comparison with the paper, use the convention that matches their tables. For physical interpretation, proper length is preferable.

Number of satellites

For a parent string segment, count the number of persistent new bulge maxima that form inside it before being absorbed by neighboring larger bulges.

Use labels:

```
n_s = integer if unambiguous,
n_s >= 1 if at least one forms but the final count is ambiguous,
? if the generation never cleanly forms before merger/absorption.
```

5.6 Plots

Produce:

1. Embedding-style snapshots:

```
R(Z) at representative times showing generations.
```

1. Growth-rate plot:

```
d ln R_i / dt versus t/r0
```

for `i=1,2,3,4` in the `L=10` case.

1. Generation table:

```
Gen, t_i/r0, t_p,i/r0, t_n,i/r0, n_s, R_s,i/r0, R_h,i/r0, L_s,i/R_s,i.
```

1. Multi-`L` table:

```
L/r0, Gen, t_p,i/r0, t_n,i/r0, n_s, R_s,i/r0, R_h,f/r0, R_s,i/L_s,i.
```

5.7 Validation criteria

For `L=10`, compare two separate tables:

```
Table-style reproduction A: threshold definition, for direct comparison with the
older Lehner–Pretorius convention.
Table-style reproduction B: derivative-based tn,i definition, for direct
comparison with the paper's preferred Table II convention.
```

The first two generations should agree with the paper at the percent level if the same definitions are used. Later generations are more ambiguous and should be treated with larger error bars.

Expected qualitative structure:

```
First generation forms first and defines the main blob/string segmentation.
Second generation forms near the center of the preceding string segment.
Third and later generations become increasingly sensitive to local dynamics.
No clean global timescale relates all later generations.
```

A successful reproduction does not require exact late-generation event times if the simulations show the same local-dynamics picture and the same lack of a universal global generation timescale.

5.8 Failure modes

- **Generations appear at asymmetric locations too early:** AMR hierarchy may be breaking `Z2` symmetry.
- **Derivative peaks are noisy:** output cadence is too low or smoothing is inadequate.
- **Bulge tracks cross or disappear:** implement track splitting/merging logic.
- **Generation table depends strongly on smoothing:** the event is physically ambiguous; report uncertainty rather than forcing a precise number.

---

6. Convergence of All Diagnostics

6.1 Target

Demonstrate that the reproduction results are not artifacts of grid spacing, AMR, horizon finding, or post-processing.

The paper’s explicit convergence test focuses on the apparent-horizon area, but for our purposes we need convergence tests for all diagnostics because the downstream gravitational-radiation project will rely on robust horizon dynamics.

6.2 Resolution set

For the primary convergence case, use `L/r0 = 10` and `r0 = 1`.

Set the medium resolution to

```
h = 0.25 r0
```

on the coarsest grid.

Use:

```
Delta_low  = (10/7) h,
Delta_med  = h,
Delta_high = (2/3) h.
```

Use the same physical setup, gauge, perturbation, boundary conditions, AMR philosophy, and output cadence across all three runs.

6.3 Expected convergence factors

Assume a diagnostic `F_h(t)` behaves as

```
F_h(t) = F_exact(t) + C h^p + higher-order terms.
```

Then the expected rescaling factor is

```
Q_p = [(Delta_low/Delta_med)^p - 1] / [1 - (Delta_high/Delta_med)^p].
```

For the chosen spacings:

```
Q_2 ~ 1.87,
Q_3 ~ 2.72,
Q_4 ~ 3.94.
```

The paper observes roughly third-order convergence for the apparent-horizon area, which is plausible even with 6th-order spatial derivatives because AMR interpolation/extrapolation and RK4 time stepping reduce the effective order.

6.4 Convergence diagnostic 1: apparent-horizon area

For each resolution, compute

```
A_low(t), A_med(t), A_high(t).
```

Interpolate all curves to common times. Then plot:

```
|A_low/A0 - A_med/A0|,
|A_med/A0 - A_high/A0|,
Q_2 |A_med/A0 - A_high/A0|,
Q_3 |A_med/A0 - A_high/A0|,
Q_4 |A_med/A0 - A_high/A0|.
```

Successful third-order convergence means the first difference is close to the `Q_3`-rescaled second difference for a significant time interval.

6.5 Convergence diagnostic 2: minimum radius

Compute

```
R_min_low(t), R_min_med(t), R_min_high(t).
```

Use the same common-time interpolation and plot the same difference structure.

Because `R_min` is a pointwise extremum, it may converge less cleanly than the area. Therefore also test convergence of:

```
z_min(t),
local quadratic fit near the minimum,
R_min after mild smoothing.
```

Successful convergence means:

```
R_min curves agree through the scaling regime,
late-time deviations shrink with resolution,
z_min tracks the same physical neck.
```

6.6 Convergence diagnostic 3: pinch-off parameters

For each resolution, fit

```
R_min = a (tc - t)
```

using identical fitting-window rules. Compare:

```
a_low, a_med, a_high,
tc_low, tc_med, tc_high,
optional exponent p_low, p_med, p_high.
```

Report convergence as:

```
Delta a_med-high,
Delta tc_med-high,
window-systematic error,
resolution-systematic error.
```

A reasonable acceptance criterion is:

```
resolution shift in tc smaller than fitting-window systematic,
resolution shift in a small compared with the L-dependence being claimed.
```

6.7 Convergence diagnostic 4: Fourier coefficients

For each resolution, compute the same segment and same coefficients:

```
c_k^low(t), c_k^med(t), c_k^high(t).
```

Compare:

```
c1/c0,
c3/c0,
c5/c0,
even-mode contamination,
least-squares residual.
```

Because coefficients require segment identification, perform two comparisons:

1. **Fixed segment convention:** use the same coordinate interval in all resolutions.
2. **Tracked segment convention:** let each resolution identify its own segment endpoints, then map to normalized segment coordinate.

Successful convergence means:

```
c1 dominance is resolution-independent,
even modes remain negligible,
relative ordering c1 >> c3 >> c5 is stable for L=10.
```

6.8 Convergence diagnostic 5: generation formation times

Generation times are event diagnostics, so they will not converge like smooth fields. Use event-time comparison:

```
t_i^low, t_i^med, t_i^high,
t_p,i^low, t_p,i^med, t_p,i^high,
t_n,i^low, t_n,i^med, t_n,i^high.
```

For the first two generations, require percent-level agreement. For later generations, report uncertainty bands and qualitative agreement.

Also compare geometric properties at the event times:

```
R_s,i,
R_h,i,
L_s,i,
n_s.
```

Use the medium-resolution event time as a common reference and evaluate all resolutions at that time as an additional comparison. This separates event-time uncertainty from radius-measurement uncertainty.

6.9 Convergence diagnostic 6: horizon reconstruction systematics

For each resolution and time, track:

```
horizon finder residual,
area estimator A versus B difference,
raw versus smoothed R_min difference,
Fourier fit residual,
```

```
constraint norm near the horizon,
number of horizon points across thinnest neck.
```

This will identify whether poor convergence is coming from the evolution itself or from post-processing.

6.10 Acceptance criteria for declaring reproduction complete

Declare the reproduction successful only when the following are true:

1. `A/A0` for `L=10` matches the benchmark curve and final value.
2. `R_min(t)` decreases toward zero and supports a stable finite-`tc` extrapolation.
3. The log-log plot has slope close to 1 in the late regime.
4. The `L=10` Fourier analysis shows `c1` dominance and negligible even modes.
5. The first two generation times and radii agree well with the paper's tables under matching definitions.
6. The area shows approximately third-order convergence for the tested interval.
7. The other diagnostics are at least qualitatively convergent, with quantified uncertainties.
8. Late-generation ambiguities are explicitly labeled rather than overfit.

---

7. Suggested Work Schedule

Phase 1: Minimal medium-resolution reproduction

1. Implement initial data for `L=10`, `r0=1`.
2. Run a short evolution to verify gauge settling and constraint damping.
3. Add apparent-horizon finder output.
4. Compute `A/A0` and `R_min/r0`.
5. Produce first versions of area and minimum-radius plots.

Deliverable:

```
L10_med preliminary reproduction notebook.
```

Phase 2: Horizon-analysis pipeline

1. Build reusable horizon post-processing library.
2. Implement extrema detection and tracking.
3. Implement generation-time definitions.
4. Implement Fourier coefficient extraction.
5. Implement pinch-off fits.

Deliverable:

```
horizon_analysis.py or equivalent,
validated on L10_med.
```

Phase 3: Full medium-resolution sweep

Run:

```
L/r0 = 8, 10, 12, 16
```

at medium resolution.

Deliverables:

```
multi-L area plot,
multi-L generation table,
multi-L pinch-off fit table,
Fourier comparison where segments are clean.
```

Phase 4: Convergence campaign

Run:

```
L10_low,
L10_med,
L10_high.
```

Deliverables:

```
area convergence plot,
R_min convergence plot,
pinch-off parameter convergence table,
Fourier convergence plot,
generation-time convergence table.
```

Phase 5: Reproduction memo

Write a short internal note with:

```
simulation parameters,
post-processing definitions,
```

```
plots,
tables,
convergence results,
known mismatches,
lessons for gravitational-radiation extraction.
```

Only after this point should the project move to wave extraction or radiation diagnostics.

---

8. Notes for the Later Radiation Project

The reproduction diagnostics here are not just bookkeeping. They are prerequisites for radiation extraction.

- The apparent-horizon area gives a rough upper bound on how much energy can have escaped as gravitational radiation, though it is not itself a waveform diagnostic.
- The generation times define the natural time windows for radiation: first-generation burst, intermediate cascade, near-pinch-off regime, and post-pinch-off collision-like dynamics.
- The Fourier coefficients identify which GL harmonics dominate the source dynamics.
- The minimum-radius scaling tells us how close to the singular local regime the simulation has reached.
- Convergence of horizon quantities is a necessary but not sufficient condition for convergence of wave extraction.

A clean next step after reproduction is to add radiation extraction at large radius while preserving all of the above diagnostics in the same run metadata.